

A STYLE MODEL INDUCTION


STYLE REFERENCE
**Small,
high-quality
dialogue corpus**

dialogue data

1

**Dialogue Feature
Analysis**
states • strategies
transitions

dialogue features

2

**Bayesian Dialogue
Modeling**
transition probabilities
observation likelihoods

B DATA SELECTION & ADAPTATION


CANDIDATE POOL
**Large,
general-purpose
dialogue corpus**

dialogue data

3

**Style Compatibility
Scoring**
 $\text{Score}(r_t) \in [0, 1]$

scores

4

**Preference Data Selection
& DPO Adaptation**
chosen $\succ$ *rejected*

LoRA-DPO


**Style-adapted
local LLM**

Bayesian dialogue model